

10. As the doping to a pure semiconductor increases, the bulk resistance of the semiconductor
- A) Remains the same B) Increases
C) Decreases D) Becomes zero
11. In a N-type semiconductor, the position of Fermi-level is
- A) Close to the valance band B) In the middle of the energy band gap
C) Can be any where D) Close to the conduction band
12. Measurement of Hall coefficient enables the determination of:
- A) Temperature coefficient and thermal conductivity B) Mobility and carrier concentration
C) Fermi level and forbidden energy gap D) Area of the device
13. The density of charge carriers in a pure semiconductor is proportional to
- A) $\exp(-E_g/kT)$ B) $\exp(-2E_g/kT)$
C) $\exp((-E_g/kT^2))$ D) $\exp(-E_g/2kT)$
14. The primary function of a solar cell is
- A) To store solar energy in chemical form B) To reflect sunlight to solar heaters
C) To convert sunlight directly into electrical energy D) To absorb and retain heat from sunlight
15. When there is no incident light, the reverse current in a photodiode is essentially non-existent and is referred to as
- A) Zener current B) Dark current
C) Photocurrent D) PIN current
16. The magnetic susceptibility (χ) in superconductor is
- A) Positive B) Zero
C) Negative D) Infinity
17. Emission of a photon by an excited atom due to interaction with a passing photon nearby is called
- A) Stimulated emission B) Induced absorption
C) Spontaneous emission D) Thermionic emission
18. The purpose of the optical cavity in a laser is to:
- A) Convert electrical energy into light B) Amplify the light through multiple passes between mirrors
C) Filter out unwanted frequencies of light D) Focus the light into a beam
19. The numerical aperture (NA) in optical fiber is used to describe
- A) Light gathering or light-collecting ability B) Light spreading ability
C) Light output from an external shield D) Light leakage ability
20. The loss in signal power as light travels down a fiber is called
- A) Dispersion B) Scattering
C) Absorption D) Attenuation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write any four differences between classical and quantum computing.	4	L*2	1	1, 2
	b) Explain the interaction of Pauli X, Y, Z matrices on $ 0\rangle$ and $ 1\rangle$ states.	8	L2	1	1, 2
	c) If $1\alpha\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$ and $1\beta\rangle = \begin{pmatrix} c \\ d \end{pmatrix}$ then prove that $\langle\alpha \beta\rangle = \langle\beta \alpha\rangle^*$	4	L3	1	1, 2
2.	a) What is a qubit? Write any four properties of qubits.	4	L2	2	1, 2
	b) Explain the operation on quantum states for the following gates with the truth table: (i) Hadamard Gate (ii) S gate	8	L2	2	1, 2
	c) $ \psi\rangle = A(2 0\rangle + 3i 1\rangle)$ find $\langle\psi \psi\rangle$	4	L3	2	1, 2

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|-------|--|---|----|---|------|
| 3. a) | Write the fundamental postulates of quantum mechanics. | 4 | L2 | 2 | 1, 2 |
| b) | What is a Bloch sphere? Explain the representation of qubits on Bloch sphere. | 8 | L2 | 2 | 1, 2 |
| c) | Verify if $ w\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ & $ v\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are orthogonal. | 4 | L3 | 2 | 1, 2 |

Unit – II

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|-------|---|---|----|---|------|
| 4. a) | What is an intrinsic semiconductor? Derive an expression for the conductivity of an intrinsic semiconductor. | 6 | L2 | 3 | 1, 2 |
| b) | Explain the effect of temperature on the Fermi level in an extrinsic semiconductor. | 6 | L2 | 3 | 1, 2 |
| c) | Mobilities of electrons and holes in a sample of intrinsic germanium at 300 K are $0.36 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$ and $0.17 \text{ m}^2 \text{ V}^{-1} \text{ S}^{-1}$ respectively. If the resistivity of the specimen is $2.12 \text{ } \Omega \text{ m}$, compute the intrinsic carrier density. | 4 | L3 | 3 | 1, 2 |
| 5. a) | Explain the construction and working of solar cell. | 4 | L2 | 3 | 1, 2 |
| b) | State Hall Effect. Explain how Hall field is set up in the given semiconductor sample and obtain an expression for the Hall voltage and Hall coefficient of an n-type semiconductor. | 8 | L2 | 3 | 1, 2 |
| c) | The Hall coefficient of a specimen of a doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3 \text{ C}^{-1}$ and its resistivity is found to be $8.93 \times 10^{-3} \text{ } \Omega \text{ m}$. Calculate charge carrier concentration and electron mobility at room temperature. | 4 | L3 | 3 | 1, 2 |
| 6. a) | What are superconductors? Explain Meissner effect in superconductors. | 4 | L2 | 4 | 1, 2 |
| b) | Explain Type-I and Type-II superconductors with suitable diagrams. | 8 | L2 | 4 | 1, 2 |
| c) | The critical temperature and critical magnetic field at 0K for superconducting lead are 7.2 K and 800 gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 gauss? | 4 | L3 | 4 | 1, 2 |

Unit – III

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|-------|--|---|----|---|------|
| 7. a) | What is a laser? Explain the principle of stimulated emission of radiation. | 4 | L2 | 5 | 1, 2 |
| b) | With neat diagrams, describe the construction and working of Nd:YAG laser. | 8 | L2 | 5 | 1, 2 |
| c) | Calculate the wavelength of emission from a GaAs semiconductor laser whose band gap energy is 1.44 eV. | 4 | L3 | 5 | 1, 2 |
| 8. a) | What is an optical fiber? Describe how the optical fiber can be used in communication system. | 4 | L2 | 5 | 1, 2 |
| b) | With diagram derive an expression for numerical aperture of an optical fiber in terms of R.I. of core and cladding. | 8 | L2 | 5 | 1, 2 |
| c) | A fiber has a core diameter of $6 \text{ } \mu\text{m}$ and its core refractive index is 1.47 and for cladding it is 1.43. How many modes can propagate into the fiber if the wavelength of the source is $1.5 \text{ } \mu\text{m}$. | 4 | L3 | 5 | 1, 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
